Guntha Sai Vedanth

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.20
Gowtham Junior College	Class XII (TSBIE)	2024	97.30%
HAL Secondary School	Class X (CBSE)	2022	92.00%

EXPERIENCE

Mechanical Design Intern at DRDL, DRDO:

(SolidWorks, ANSYS)

- **Designed** a passive **retention mechanism** to **securely hold** a tube-launched missile **in position** until launch.
- **Evaluated** alternative design concepts while considering **functionality**, **spatial constraints**, and **component availability**.
- **Refined CAD models** through **iterative design improvements** to optimize size, weight, and integration.
- **Ensured seamless integration** of the mechanism with the **existing system** while **satisfying assembly requirements**.

PROJECTS

Planar Kinematic Mechanism Simulator:

- **Developed** a **generalized** web-based simulation platform for **any arbitrary 1-DoF planar kinematic mechanisms**.
- **Designed** a reusable **object-oriented** framework using **modular link and joint components** to enable **mechanism-independent** analysis and **eliminate mechanism-specific implementations**.
- **Built** a generalized **Newton-Raphson** solver with **Jacobian-based** formulations for **kinematic analysis**.
- **Enabled** computation of **reaction forces, moments, and input forces** through a **dynamic force analysis** module.

Cost-effective Prosthetic Arm (Ongoing)**:**

- **Developing** an **affordable, fully mechanical** prosthetic arm designed to improve accessibility for **cost-sensitive users**.
- **Designed** a **spring-actuated mechanism** driven by **shoulder movement**, eliminating the reliance on electronics.
- **Applied static balancing** techniques to **reduce the effort required** from the user to open the prosthetic hand.
- **Developing** a **switching mechanism** to transition between **low-effort opening and high-force gripping** modes.

Finite Element Analysis (FEA) Simulator:

- **Engineered** a **modular OOP FEM solver** supporting **5 element types** across **2D, 3D solid, and truss formulations**.
- Built an **Abaqus .inp preprocessing pipeline** for nodes, elements, sets, forces, pressures, and boundary conditions.
- Implemented **global stiffness-matrix assembly**, solved **K·u = f**, and developed **post-processing** for structural analysis.
- **Validated** solver accuracy against **Abaqus benchmarks**, achieving **<5% deviation** across test cases.

Cycle Theft Detection System:

- **Designed** and **developed** an anti-theft system for bicycle using **Arduino**, **MPU sensors**, and **Bluetooth** communication.
- Implemented **real-time motion detection** to identify unauthorized bicycle movement and potential theft attempts.
- **Developed** a mechanism that **triggers** an audible **alarm** to **deter theft** and **alert** nearby individuals.

SKILLS

CAD Softwares: SolidWorks, Solid Edge, Fusion 360, Onshape, Creo

Simulation Tools: ANSYS, SIMULIA Abaqus, Simulink, MSC Adams

Programming Languages: MATLAB, C, Python, TypeScript (JavaScript)

SCHOLASTIC ACHIEVEMENTS

Academic Excellence Award, IIT Hyderabad**:**

- **Awarded for securing the highest GPA** among all the eligible students in the class across the **2nd and 3rd semesters**.

POSITIONS OF RESPONSIBILITY

Community Domain Head of Cepheid - Astronomy and Astrophysics club of IITH:

- **Managed** recruitment and selection of club members, helping **build** and **guide** the club's team.
- **Led** and **coordinated** the club's core team, **overseeing** the **planning** and **execution** of the club activities.

Core member of Cepheid - Astronomy and Astrophysics club of IITH:

- **Conducted** Community Sessions on astronomy, **fostering scientific curiosity** and **engagement** among IITH community.
- **Organized** telescope observation events for **150+ attendees**, providing exposure to observational astronomy.

RELEVANT COURSES

Relevant Mechanical Core Courses:

Design of Machine Elements, Kinematics and Dynamics of Mechanisms, Fundamentals of Robotics, Mechanics of Solids, Engineering Drawing, FEM and CFD Theory, Digital Fabrication, Manufacturing Science-I & II

Relevant Course on Engineering Simulations:

A Hands-on Introduction to Engineering Simulations, FEM lab, CFD lab